V.L Memorial Public School

475 Vijay Nagar

CHAPTER-12-QUESTIONS-CLASS8TH-SCIENCE

Class 08 - Science (Curiosity)

Section A

1. What can excessive use of synthetic fertilisers reduce in agricultural soil? **[1]**

   a) Water availability
   b) Forest cover
   c) Soil fertility
   d) Crop diversity

2. What do biotic components in an ecosystem rely on for nutrition, respiration, and reproduction? **[1]**

   a) Human-made substances
   b) Abiotic components
   c) Other biotic components
   d) Artificial environments

3. How do plant roots affect the surrounding soil? **[1]**

   a) They remove sunlight from soil
   b) They increase soil erosion
   c) They convert soil into water
   d) They hold soil in place

4. Which two types of components are typically found in a habitat? **[1]**

   a) Mechanical and Physical Components
   b) Automated and Digital Components
   c) Chemical and Fossil Components
   d) Biotic and Abiotic Components

5. Which type of forest is found extensively in the Sundarbans? **[1]**

   a) Mangrove forests
   b) Deciduous forests
   c) Coniferous forests
   d) Thorn forests

6. Which of the following is a human-made ecosystem? **[1]**

   a) Natural forest
   b) Fish pond
   c) Wild grassland
   d) Natural wetland

7. What do elephants search for when they enter farms and villages due to habitat changes? **[1]**

   a) Shelter and warmth
   b) Food like bananas and sugarcane
   c) Companionship with domestic animals
   d) Predators

8. Which ecosystem provides resources such as timber, fibres, medicines, and fertile soil? **[1]**

   a) Forest ecosystem
   b) Coastal ecosystem
   c) Aquatic ecosystem
   d) Artificial ecosystem

9. Which organisms form the biotic components of a habitat? **[1]**

a) Plants, animals, and microorganisms
b) Air
c) Sunlight
d) Water and soil

10. Why is it important for different groups of organisms to live together in a habitat? **[1]**

a) To share resources for survival
b) To prevent overpopulation of one type
c) To create competition
d) To eliminate predators

11. Which organism occupies the second trophic level in a food chain? **[1]**

a) Large carnivores
b) Small carnivores
c) Herbivores
d) Green plants

12. Why do different organisms live in different habitats? **[1]**

a) They depend only on sunlight
b) They require identical surroundings
c) They avoid every physical condition
d) They need specific survival conditions

13. Which organisms are examples of omnivores? **[1]**

a) Deer and hare
b) Fish and insects
c) Crows and foxes
d) Frogs and leopards

14. Which organism is a biotic component of a pond habitat? **[1]**

a) Water
b) Temperature
c) Sunlight
d) Fish

15. What is the primary diet of a herbivore in a grassland ecosystem? **[1]**

a) Meat
b) Insects
c) Plants
d) Fish

16. Why are plants called producers? **[1]**

a) They make their own food
b) They eat other organisms
c) They depend completely on soil
d) They obtain food from animals

17. Which flower part releases pollen grains? **[1]**

a) Male reproductive stamens
b) Female reproductive carpels
c) Coloured flower petals
d) Green leafy sepals

18. Which pair correctly represents one biotic component and one abiotic component of a pond? **[1]**

a) Water and dragonfly
b) Temperature and frog
c) Fish and water
d) Sunlight and lotus

19. What are the non-living components of a habitat called? **[1]**

a) Biotic components
b) Animal communities
c) Living organisms
d) Abiotic components

20. Which national park is located in Manipur? **[1]**

a) Jim Corbett National Park
b) Keibul Lamjao National Park
c) Manas National Park
d) Hemis National Park

21. **Assertion (A):** Fish in a pond influence the number of flowering plants nearby by affecting the population of pollinating insects. **[1]**
Reason (R): Fish prey on dragonfly larvae, reducing the number of dragonflies that prey on pollinators like bees and butterflies.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

22. **Assertion (A):** Competition among organisms in an ecosystem helps control population size and maintain ecosystem balance. **[1]**
Reason (R): Without competition, one species might multiply excessively, leading to ecological imbalance.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

23. **Assertion (A):** The increase in insect population in ponds is a result of decreased fish numbers. **[1]**
Reason (R): Fish consume insects, and fewer fish lead to an increase in insects.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

24. **Assertion (A):** Changes in ecosystems can be human-made or natural. **[1]**
Reason (R): Ecosystem balance is dynamic and requires stable interactions among organisms

a) Both A and R are true, and R is the correct explanation of A.
b) Both A and R are true, but R is not the correct explanation of A.
c) A is true, but R is false.
d) A is false, but R is true.

25. **Assertion (A):** The removal of frogs from a food chain can increase the grasshopper population. **[1]**
Reason (R): Frogs are natural predators of grasshoppers.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

26. **Assertion (A):** Synthetic fertilizers improve soil health in the long term. **[1]**

Reason (R): They reduce the population of beneficial microorganisms in the soil.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

27. **Assertion (A):** Mushrooms growing on dead plants help in the decomposition process. **[1]**
Reason (R): Fungi break down complex substances in dead matter into simpler nutrients, enhancing soil fertility.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

28. How do living organisms coexist harmoniously in the same habitat? **[1]**
29. How does cutting down trees for construction affect wildlife habitats? **[1]**
30. What are biotic components of a habitat? Give examples of living beings that form these components within that habitat. **[1]**
31. How do fish obtain food and oxygen in a pond? **[1]**
32. Define population in a habitat. **[1]**
33. What is mutualism? **[1]**
34. How do organisms interact with abiotic components? **[1]**

Section B

35. What is a food chain, and how does it represent ecological relationships within an ecosystem? **[2]**
36. Describe how a pond provides essential conditions for fish to survive. **[2]**
37. Explain the meaning of community in a habitat. **[2]**
38. Interpret the importance of observing multiple types of organisms in a habitat. **[2]**
39. How can human actions disturb ecological balance? **[2]**
40. How do wildlife corridors help elephants move safely between different forest areas? **[2]**
41. What role do biotic components play in a community within a habitat? **[2]**

Section C

42. Describe the function of decomposers in nutrient cycling. **[3]**
43. List the components required for producing food in plants. **[3]**
44. Explain why migratory birds are important for ecosystems during their journeys. **[3]**
45. Compare two habitats based on their common characteristics and differences in components. **[3]**
46. How do decomposers contribute to the concept of nature's recycling process? **[3]**
47. Discuss the potential impact of human-driven activities such as overfishing on the balance between biotic and abiotic components within ecosystem communities. **[3]**
48. Explain the significance of different organisms living together in a habitat. **[3]**

Section D

49. **Read the following text carefully and answer the questions that follow:** **[4]**
A study compared two ponds-one with fish and one without. Pond A (with fish) had fewer dragonflies because fish ate their larvae. As dragonflies decreased, more bees and

butterflies thrived, increasing pollination and seed production in nearby plants. This shows the interdependence of organisms, where fish indirectly support plants through pollinators. It highlights how biotic and abiotic factors interact and how human activities like overfishing may disturb ecological balance.

Questions:

a. Which factor led to fewer dragonflies in Pond A? **(1)**

b. How does the presence of fish in a pond indirectly affect seed production in nearby plants? **(1)**

c. Explain how the study of ponds shows the interdependence of organisms in a community. **(2)**

OR

How might human activities like overfishing disturb the balance between living and non-living components of an ecosystem? **(2)**

50. **Read the following text carefully and answer the questions that follow:** **[4]**

Elephants often move into farms when forests shrink due to deforestation, scarce rainfall, and rising temperatures. Loss of vegetation and drying waterholes make survival in forests difficult. Wildlife corridors have been created to connect fragmented forests and reduce conflict with humans. Every habitat has biotic and abiotic components. Organisms depend on these resources and adapt differently-like fish in ponds or snakes and rodents in forests-showing interconnections between living beings and their environment.

Questions:

a. Name an abiotic component of a habitat. **(1)**

b. Why do elephants often enter human settlements and farms? **(1)**

c. How do fish survive in a pond habitat? **(2)**

OR

Explain how two organisms in the same habitat may experience different conditions. **(2)**

51. **Read the following text carefully and answer the questions that follow:** **[4]**

In an ecosystem, organisms are linked through feeding relationships that form food chains and food webs. A food chain shows "who eats whom," such as grass $\longrightarrow$ grasshopper $\longrightarrow$ frog $\longrightarrow$ snake $\longrightarrow$ eagle. Each level in the chain is called a trophic level, with producers at the base and carnivores at higher levels. Multiple food chains often interconnect, forming a food web that illustrates the complex interdependence of organisms within ecosystems.

Questions:

a. What is the correct order of trophic levels in a simple food chain? **(1)**

b. What is a food chain, and how does it represent relationships in an ecosystem? **(1)**

c. Differentiate between a food chain and a food web. **(2)**

OR

Why are producers considered the foundation of all food chains? **(2)**

Section E

52. Analyze what could happen if a habitat contained only one type of living organism. **[5]**

53. What common challenges do elephants face when their natural forest habitats are diminished, and how do these challenges relate to the concept of habitat interdependence? **[5]**

54. Identify and explain two ways in which human activities threaten the Sundarbans ecosystem. **[5]**

55. Define the term "population" in the context of ecological studies and provide an example to illustrate your explanation. **[5]**

56. Identify and describe the three main types of interactions between organisms in an ecosystem that maintain balance. **[5]**

Section F

57. **State whether the given statement is True or False:** **[9]**

(a) The grasshopper is part of the food chain involving grass, frogs, snakes, and eagles. **[1]**

(b) In a grassland ecosystem, the hare is eaten by the frog. **[1]**

(c) Mangroves can help reduce the impact of tsunamis on coastal villages. **[1]**

(d) Food chains can include more than one type of organism at each level. **[1]**

(e) In an ecosystem, food chains are separate and do not interlink. **[1]**

(f) Frog populations remain unaffected by environmental changes. **[1]**

(g) Fish have no role in maintaining the ecological balance in a pond community. **[1]**

(h) In mutualism, only one organism benefits while the other is harmed. **[1]**

(i) Herbivores occupy the top trophic level in a food chain. **[1]**

Section G

58. **Fill in the blanks:** **[10]**

(a) In an ecosystem, organisms that consume other organisms are known as _____. **[1]**

(b) The interaction and dependence among organisms in a habitat form a _____. **[1]**

(c) The relationship where one organism benefits and the other is harmed is called _____. **[1]**

(d) A group of fish of the same kind living together is called a _____. **[1]**

(e) A population consists of organisms of the same type living in the same _____. **[1]**

(f) Organisms like fungi and bacteria that break down dead matter are called _____. **[1]**

(g) Fungi and bacteria are microorganisms that break down complex substances into simpler ones through a process called _____. **[1]**

(h) The reproductive parts of a flower, stamens and carpels, are crucial for the process of _____. **[1]**

(i) Forests and grasslands are examples of _____ ecosystems. **[1]**

(j) The process in which plants release oxygen into the atmosphere is called _____. **[1]**